
Berkeley, CA | oliverhchang@berkeley.edu | (408) 203-9356

EDUCATION

University of California, Berkeley

Class of 2028

B.S. Mechanical Engineering

EXPERIENCE

Formula Electric at Berkeley

Sept 2024 - Present

Chassis Mechanical Engineer

- Optimization of spaceframe strength and weight through structural analysis and impact testing using ANSYS Workbench, Spaceclaim, and Mechanical, informing rule compliant chassis geometry and improved steering
- Validated rule compliance and assessed Factor of Safety under crash, cornering, and braking scenarios
- Designed torsional stiffness testing jig and testing plan to validate simulation data and inform handling performance

UC Berkeley's Space Technology and Rocketry (STAR)

Sept 2024 - Present

Airframe & Payload Engineer

- Designed K-Class rocket layout on OpenRocket, conducted CFD simulations, fabricated airframe and recovery parts with a group of 8 all within a single semester
- Assembled, soldered, and programmed custom avionic and payload boards for altitude, pressure, and temperature data
- Led full development process of custom insulated Arduino-powered ice-cream maker payload, utilizing rapid prototyping and 3D printing to create a fully-functioning device over 3 iterations

Fremont High Robotics FRC Team 3501: Firebots

2020 - 2024

VP of Mechanical Design

- Led a team of 6 to design a competition robot for the 2024 FIRST Robotics Competition. The first iteration was designed within the first 2 weeks of build season, with an overhaul of the intake and flywheels after the first competition
- Coordinated with manufacturing to create industry-standard drawings & deliver high-quality parts
- Redesigned school workshop by designing and building 13 custom tables, 2 CNC Enclosures, with overhauled of tool and machine organization to streamline work processes
- Finished top 15% of all teams in the world

Business Lead

- Led a team to apply for 15+ grants, overhauled grant writing and budgeting systems from the ground-up
- Planned and hosted 35+ outreach events exposing 10,000+ attendees to FIRST opportunities
- Created 5 unique lesson plans to introduce STEAM to elementary school students, now taught monthly to 500+ students across 6 local elementary school after-school programs

WORK EXPERIENCE

Scouts BSA Summer Camp Staff

June - July 2022, 2023, 2024

Kitchen Staff, Aquatics Staff, Handicraft Director

- Designed and taught custom Handicraft curriculum for Cub Scout utilizing leatherworking & woodworking skills
- Instructed over 400+ scouts on first aid, swimming, and lifesaving merit badges

Honors & Awards

Dean's List Semifinalist, FIRST Robotics

Eagle Scout, Scouts BSA: Drafted, financed, and built originally designed equipment shed for Sunnyvale Little League, working with local suppliers to reduce costs

Skills

CAD/CAM: Onshape, Solidworks, Fusion 360, AutoCAD, ANSYS Workbench, ANSYS Mechanical, ANSYS Spaceclaim

Programming: MATLAB, Python, Java, Arduino

Manufacturing: 3D Printing, BambuStudio, CNC, Fablight, Laser Cutting, Lathe Machines, Milling Machines, Waterjet